

Experiment No.	3
Experiment Title	Loan Amount Prediction using Linear Regression ¹
Student Name	Preethi D
Register Number	3122247001046
Faculty	Dr. Poreddy Ajay Kumar Reddy
Submission Date	28-07-26

1 Objective

To predict the loan sanction amount for applicants using Linear Regression, and to compare it against regularized alternatives (Ridge, Lasso, and ElasticNet) in terms of prediction error and computational cost.

2 Problem Statement

Given a dataset of loan applicants with demographic, financial, and property-related attributes, the objective is to predict **Loan Sanction Amount (USD)**, a continuous target, making this a regression problem. Input: 23 applicant/property features (after removing the target). Output: predicted loan sanction amount in USD. Models are evaluated using MAE, MSE, RMSE, and R^2 .

3 Dataset Description

Dataset Name	Loan Amount Prediction (train.csv)
Dataset Source	Kaggle / Analytics Vidhya Loan Prediction dataset
Number of Samples	30,000
Number of Features	24 columns total (23 predictors after removing the target) – mix of numeric (Age, Income, Credit Score, etc.) and categorical (Gender, Profession, Property Location, etc.)
Number of Classes	N/A (regression task)
Missing Values	Present in 10 columns, e.g. Income (USD): 4,576; Type of Employment: 7,270; Property Age: 4,850; Credit Score: 1,703; target itself: 340
Train-Test Split	80% train / 20% test, giving 24,000 training and 6,000 test samples

¹GitHub Repository: <https://github.com/preethi340/Machine-Learning-.git>

4 Exploratory Data Analysis

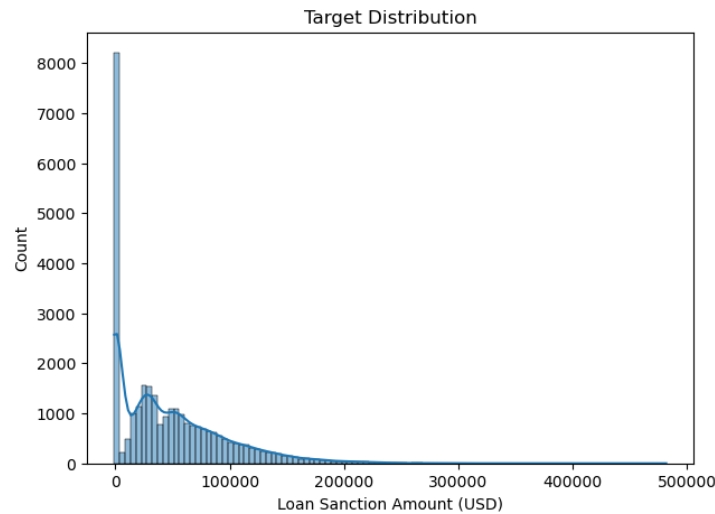


Figure 1: Distribution of the target variable, Loan Sanction Amount (USD)

Inference: The target is right-skewed, with most sanctioned loans clustering at lower-to-moderate amounts and a long tail of larger sanctioned loans extending the distribution. This skew suggests that a plain linear model may be biased toward underpredicting the largest loan amounts, and that a log-transform of the target could be worth exploring in future iterations.

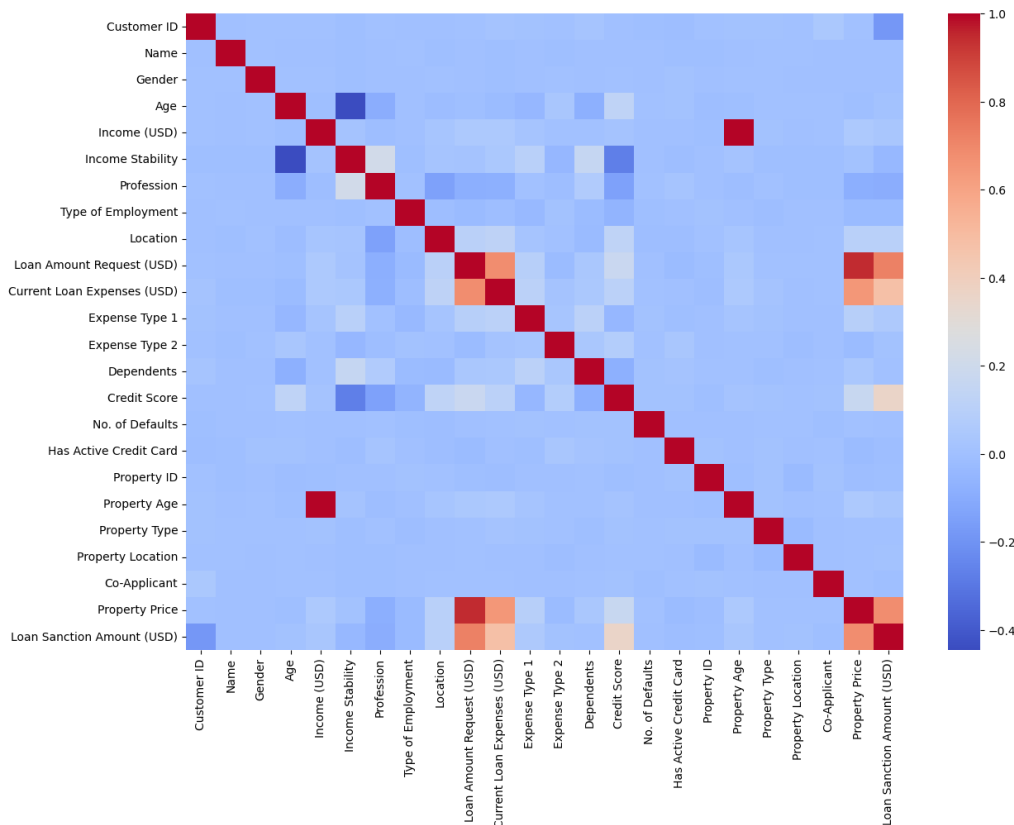


Figure 2: Correlation heatmap of all (label-encoded) dataset features

Inference: Loan Amount Request (USD) and Property Price show the strongest positive correlation with the target, Loan Sanction Amount (USD), which makes intuitive sense since sanctioned amounts are naturally tied to the amount requested and the value of the property being financed. Most demographic and identifier-like columns (Customer ID, Name, Property ID) show negligible correlation with the target, as expected, since they carry no real predictive signal.

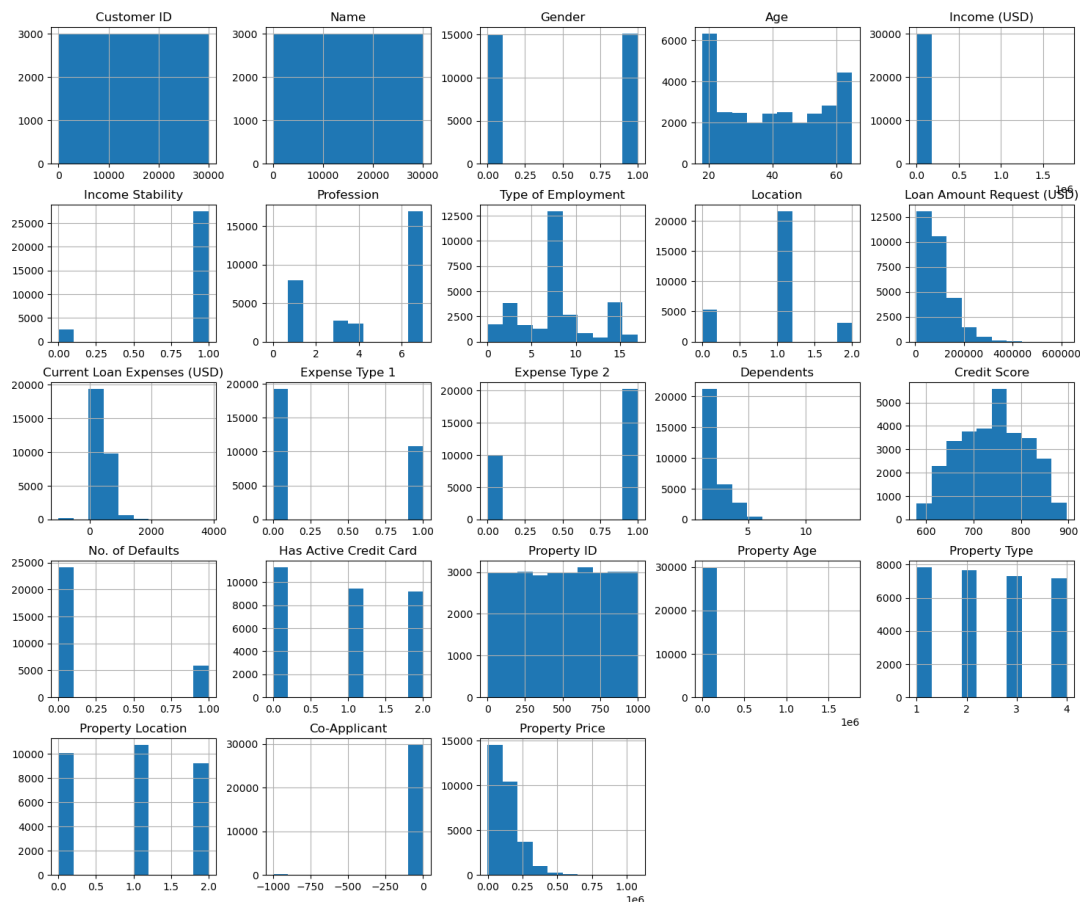


Figure 3: Distributions of all numeric/encoded features

Inference: Financial features such as Income (USD), Loan Amount Request (USD), and Property Price are all heavily right-skewed with long tails of high-value outliers, while Age and Credit Score are closer to symmetric/bell-shaped. This mix of distributions motivates the outlier-aware and scale-invariant preprocessing steps applied later (standardization, and comparing regularized regressors that are more robust to skewed inputs).

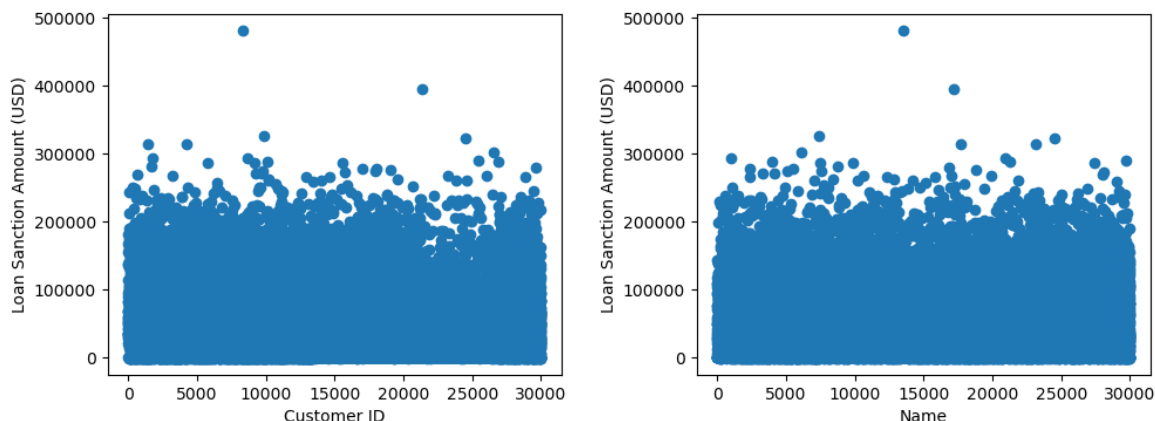


Figure 4: Scatter plots of Loan Sanction Amount vs. Customer ID (left) and Name (right), both label-encoded

Inference: As expected, the label-encoded Customer ID and Name columns show no discernible pattern or trend against the target – these are identifier-like fields that were included among the first four numeric columns purely due to column ordering, and carry no genuine predictive relationship with loan amount.

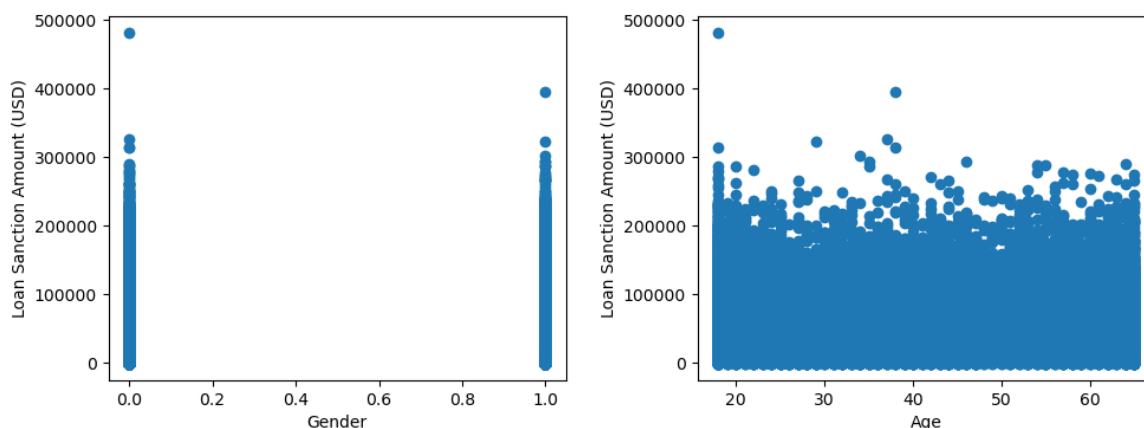


Figure 5: Scatter plots of Loan Sanction Amount vs. Gender (left) and Age (right)

Inference: Gender, being a binary encoded categorical variable, naturally forms two vertical bands with substantial overlap in loan amount between them, indicating little difference in sanctioned amount by gender alone. Age shows a wide, fairly uniform spread of loan amounts at every age value, suggesting Age alone is only weakly predictive of the sanctioned loan amount without being combined with financial features.

5 Data Preprocessing

- **Missing value handling:** Numeric columns were imputed with the column median and categorical columns with the most frequent category, both using `SimpleImputer`, reducing total missing values across the dataset from several thousand to zero.
- **Encoding:** All categorical columns (Gender, Profession, Type of Employment, Location, Expense Type 1/2, Has Active Credit Card, Property Location, plus

identifier-like Customer ID and Name) were converted to numeric codes using `LabelEncoder`.

- **Feature scaling:** All features were standardized using `StandardScaler` (zero mean, unit variance) prior to model training.
- **Train-test split:** An 80/20 split was used (`train_test_split, random_state=42`), giving 24,000 training and 6,000 test samples.
- **Feature engineering:** None beyond encoding and scaling; all 23 available columns (including identifier columns such as Customer ID and Property ID) were retained as model inputs.

6 Machine Learning Algorithm

Linear Regression (and regularized variants: Ridge, Lasso, ElasticNet):

- **Working principle:** Linear Regression models the target as a weighted linear combination of input features, learning coefficients that minimize the sum of squared residuals between predicted and actual values. Ridge adds an L_2 penalty on the coefficients to shrink them and reduce overfitting/multicollinearity sensitivity; Lasso adds an L_1 penalty that can shrink some coefficients exactly to zero (implicit feature selection); ElasticNet combines both L_1 and L_2 penalties.
- **Mathematical formulation:** Ordinary Least Squares minimizes $\sum_{i=1}^n (y_i - \hat{y}_i)^2$ where $\hat{y}_i = \beta_0 + \sum_{j=1}^p \beta_j x_{ij}$. Ridge minimizes $\sum_i (y_i - \hat{y}_i)^2 + \alpha \sum_j \beta_j^2$; Lasso minimizes $\sum_i (y_i - \hat{y}_i)^2 + \alpha \sum_j |\beta_j|$; ElasticNet minimizes $\sum_i (y_i - \hat{y}_i)^2 + \alpha \left(l1_ratio \sum_j |\beta_j| + \frac{1-l1_ratio}{2} \sum_j \beta_j^2 \right)$.
- **Advantages:** Linear Regression is simple, fast to train, easy to interpret via its coefficients, and provides a strong, transparent baseline. Regularized variants help control overfitting and multicollinearity, with Lasso additionally offering automatic feature selection.
- **Limitations:** Linear Regression assumes a linear relationship between features and target and is sensitive to outliers and skewed distributions (both observed in this dataset's financial features), which likely limits achievable accuracy; it also cannot capture non-linear interactions between features such as income and credit score without explicit feature engineering.

7 Experimental Setup

Python Version	
Scikit-Learn Version	
IDE	Jupyter Notebook
Operating System	Windows
Hardware	

8 Hyperparameter Selection

Parameter	Value / Search Space
Ridge α (GridSearchCV, 5-fold, scoring=r2)	searched {0.01, 0.1, 1, 10, 100}; best = 0.01
Lasso α (GridSearchCV, 5-fold, scoring=r2)	searched {0.001, 0.01, 0.1, 1, 10}; best = 0.001
ElasticNet α / l1_ratio (GridSearchCV, 5-fold)	$\alpha \in \{0.01, 0.1, 1, 10\}$, l1_ratio $\in \{0.2, 0.5, 0.8\}$; best =

9 Results

Single train-test split (6,000 test samples):

Model	MAE	MSE	RMSE	R ²	Training Time (s)
Linear Regression	20,785.52	9.616×10^8	31,009.45	0.5820	0.1406
Ridge	20,785.49	9.616×10^8	31,009.22	0.5820	0.3290
Lasso	20,785.51	9.616×10^8	31,009.44	0.5820	12.4710
ElasticNet	20,790.81	9.500×10^8	30,822.31	0.5871	2.5035

5-Fold Cross-Validation (on full standardized dataset):

Model	MAE	MSE	RMSE	R ²
Linear Regression	20,893.64	9.103×10^8	30,171.33	0.6040
Ridge	20,893.63	9.103×10^8	30,171.28	0.6040
Lasso	20,893.28	9.103×10^8	30,171.18	0.6040
ElasticNet	20,900.94	9.085×10^8	30,141.98	0.6048

10 Result Visualization

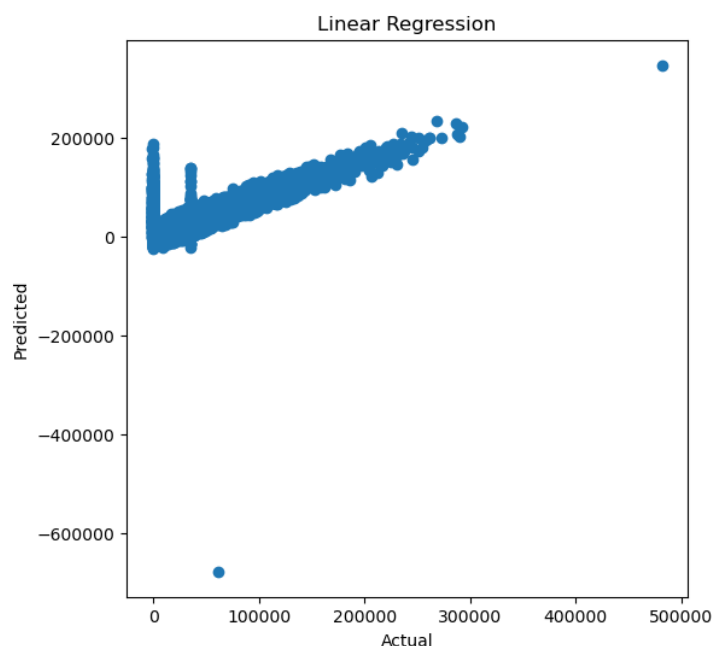


Figure 6: Actual vs. Predicted Loan Sanction Amount (Linear Regression, test set)

Inference: Points broadly follow the diagonal (perfect-prediction) line for small-to-moderate loan amounts, but the model increasingly underpredicts as the actual sanctioned amount grows, consistent with the right-skewed target distribution observed earlier. This pattern is typical of linear models applied to skewed financial targets without a log-transform.

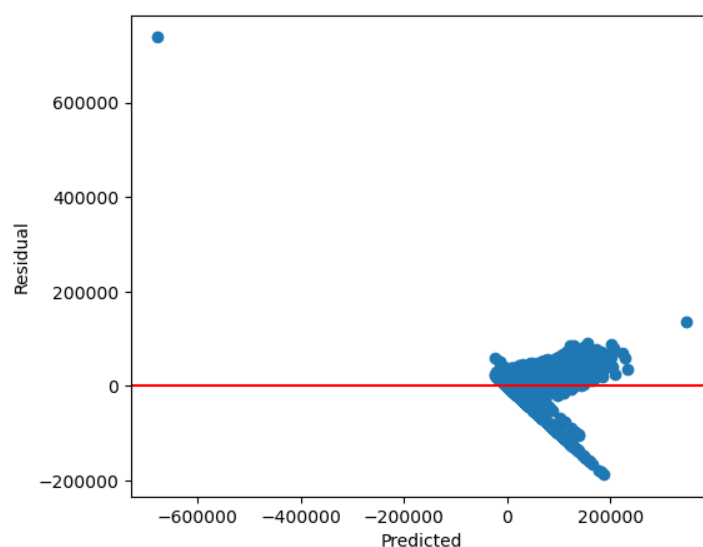


Figure 7: Residual plot (Predicted vs. Residual) for Linear Regression

Inference: Residuals fan out and grow more strongly positive as the predicted value increases, indicating heteroscedasticity (non-constant error variance) rather than the ideal

random scatter around zero. This systematic pattern suggests the linear model is missing some non-linear structure and/or would benefit from a target transformation.

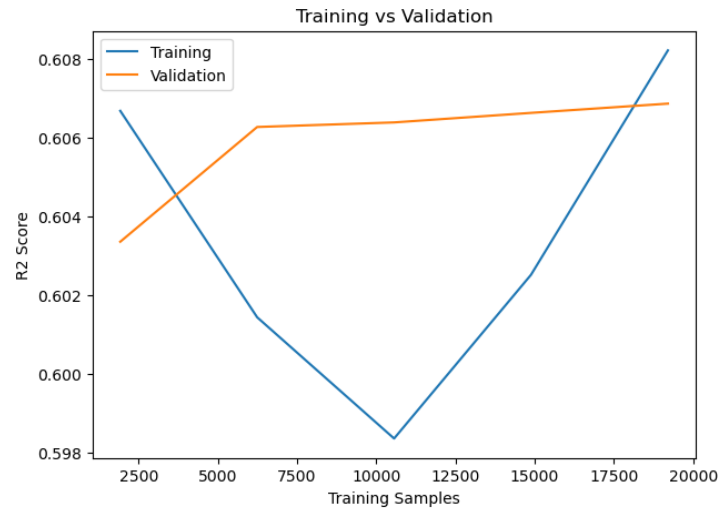


Figure 8: Learning curve: training vs. validation R^2 score as training size increases (Linear Regression)

Inference: The training and validation R^2 scores converge closely as the training set grows, with a fairly small persistent gap between them. This indicates the model has fairly high bias (underfitting) rather than high variance – adding more training data alone is unlikely to substantially improve performance; richer features or a non-linear model would likely help more.

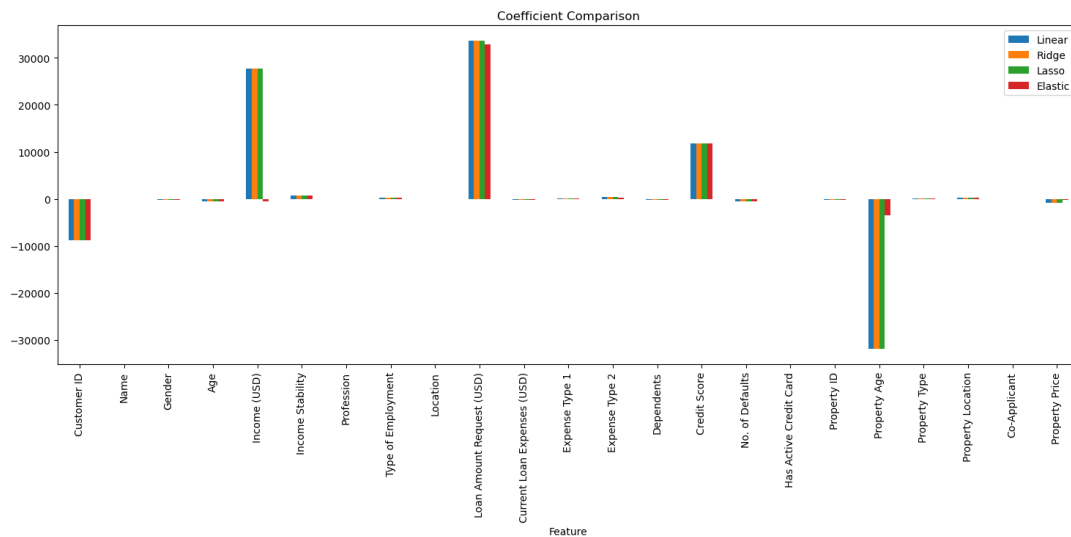


Figure 9: Coefficient comparison across Linear, Ridge, Lasso, and ElasticNet models

Inference: Linear, Ridge, and Lasso produce very similar coefficient magnitudes for most features (since their optimal regularization strengths were selected as very small), while ElasticNet – with its higher effective regularization – visibly shrinks several coefficients, most notably reducing the large **Income (USD)** coefficient seen in the other three models. This matches ElasticNet achieving the best R^2 /RMSE among the four models, suggesting mild regularization does help generalization slightly on this dataset.

11 Discussion

All four regression models achieved very similar performance, with R^2 around 0.58–0.59 on the held-out test set and around 0.60 under 5-fold cross-validation. ElasticNet was consistently the best performer by a small margin (lowest RMSE, highest R^2) once regularization strength was tuned via GridSearchCV, while Ridge and Lasso essentially reproduced Linear Regression’s performance because their tuned α values were very small (0.01 and 0.001 respectively), indicating limited multicollinearity/overfitting to correct for in this dataset. Lasso’s training time (12.47 s) was far higher than the other three (all under 0.4 s), consistent with its iterative coordinate-descent optimization being more expensive than the closed-form/simpler solvers used by the others. The residual and actual-vs-predicted plots both point to the same underlying limitation: the model systematically underpredicts large loan amounts, likely because the target and several key financial predictors are strongly right-skewed. A log-transform of the target and/or of skewed features, along with dropping non-informative identifier columns (Customer ID, Name, Property ID), would be reasonable next steps to improve R^2 beyond the current 0.60 ceiling.

12 Conclusion

This experiment applied Linear Regression, along with Ridge, Lasso, and ElasticNet, to predict loan sanction amounts from applicant and property data. After hyperparameter tuning via GridSearchCV, ElasticNet ($\alpha = 0.01$, l1_ratio=0.8) gave the best overall result (test $R^2 = 0.587$, cross-validated $R^2 = 0.605$), narrowly outperforming plain Linear Regression, while diagnostic plots (residuals, learning curve) indicate the models are underfitting somewhat, most likely due to the skewed nature of the financial features and the presence of uninformative identifier columns in the input.

13 References

1. F. Pedregosa *et al.*, “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
2. A. E. Hoerl and R. W. Kennard, “Ridge Regression: Biased Estimation for Nonorthogonal Problems,” *Technometrics*, vol. 12, no. 1, pp. 55–67, 1970.
3. R. Tibshirani, “Regression Shrinkage and Selection via the Lasso,” *Journal of the Royal Statistical Society, Series B*, vol. 58, no. 1, pp. 267–288, 1996.
4. H. Zou and T. Hastie, “Regularization and Variable Selection via the Elastic Net,” *Journal of the Royal Statistical Society, Series B*, vol. 67, no. 2, pp. 301–320, 2005.